

PetTodo — Progress Report

Abstract

PetTodo is a dual-platform (iOS and Android) mobile application pairing a deliberately minimal to-do list with a virtual pet generated from a photograph of the user's own animal, aimed at adults with ADHD or ADHD-like executive-function difficulty. Its design thesis is that gentle reminders do not initiate action in this population, whereas external anchors — dependants, animals, other people — might, so the pet digitises a mechanism the population already uses. I tested that thesis against four bodies of evidence gathered for this report, and the most consequential finding is adverse to it: the anchor mechanism turns out to be the least evidenced proposition in the design, while the reminder-and-planning approach it rejects has controlled effects in diagnosed ADHD samples. The safety architecture fared better, though the assumption that removing punishment removes guilt is contradicted by studies of trackers with no character and no failure state at all. At mid-term the core loop is functionally complete on both platforms with 47 automated tests passing; the remaining work is an evaluation designed to test the mechanism rather than assume it.

1. Introduction

1.1 Context and motivation

Conventional task-management software assumes the user can maintain the system it provides. For adults with ADHD this inverts the problem: the executive function needed to keep a productivity tool current is precisely what is impaired. The specific capacity involved is prospective memory — remembering to carry out an intention at the right moment — and it fails asymmetrically: in a case-control experiment, 25 adults with diagnosed ADHD showed a large impairment in *time-based* prospective memory while performing comparably to matched controls on *event-based* tasks (Altgassen, Kretschmer & Kliegel 2014, *Journal of Attention Disorders*). The population is also deeply heterogeneous: across six neuropsychological domains, no single deficit is present in more than a minority of diagnosed individuals — 18.1% to 36.1% depending on domain (Coghill et al. 2014, *Psychological Medicine*). Any mechanism a tool builds will therefore reach a subgroup, not the population — including the reminder mechanisms this project set out to improve upon.

The motivating observation is that the interventions this population itself reports as working are concrete and external — a child, a dog that must be walked, an alarm clock placed across the room — rather than motivational messaging. PetTodo asks whether that observation survives being built: whether a virtual pet derived from the user's own animal can serve as an external anchor for small daily tasks.

1.2 Aims and objectives

Build and evaluate a to-do application whose engagement mechanism is an external anchor rather than a reminder, without importing the guilt that has caused comparable applications to be abandoned. Objectives: (i) a dual-platform core loop working entirely offline, with no account and no server; (ii) a pet generated from the user's *own* animal, on the hypothesis that attachment to a real, recognisable companion exceeds attachment to a generic avatar; (iii) safety red lines — no punishment, no streaks, no failure states — enforced structurally rather than by discipline; (iv) an evaluation against a retention criterion, with the criterion itself examined rather than assumed.

1.3 Background

1.3.1 *The problem: abandonment, not absence of features*

The category's failure mode is documented best by its own users. Reading high-engagement threads across r/ADHD, r/adhdwomen and r/ADHD_Programmers — including a 4,625-upvote account of 500 days spent trialling 36 productivity applications — three patterns recur. Feature completeness does not solve the problem: the highest-voted example describes a user who built himself an application that ingested his email, prioritised automatically and contacted him each morning, and still did not use it; friction is not the barrier, initiation is. Second, maintaining the tool is itself an executive-function task, so the tool decays exactly when the user does. Third, the community proposes its own evaluation standard: early reviews are worthless because ADHD reviewers are novelty-seeking by disposition, and only two-week retention convinces. A serial abandoner's description supplies the phrase this project now designs against: an app that "feels like my disappointed mother", where the user feels "guilty about the app I downloaded to stop feeling guilty about tasks" — a state commenters name a *shame reminder*.

1.3.2 *Survey of existing systems*

I surveyed the four dominant applications in this space through 4,754 store reviews collected for this report (method and sampling caveats in §2.2), plus their public store metadata.

Finch (virtual bird plus self-care tasks; 739K App Store ratings at 4.9 ★, 10M+ Play installs) is the category leader and the reference product for the pet mechanism. Emotional-bond language ("my birb", "companion", "attached") appears unprompted in 29% of its recent Apple reviews, against 9–13% for the other three — the clearest external evidence that the pet mechanism does something the others do not. Its user-reported failure modes are equally instructive: guilt on lapse severe enough that users stop opening the app; falsified completions to earn pet rewards, recognised as self-deception while it happens; an aesthetic that adults repeatedly call infantile; and — its top-voted Android complaint — data loss, with multi-year pets wiped and users explicitly requesting a manual backup button the product does not provide. One detailed user account also locates the mechanism's effective range: the pet reward does not motivate large goals ("being rewarded by buying clothes for a virtual bird isn't enough motivation"), but works for small daily maintenance — finishing a bottle of water, stretching, taking medication — and the app is opened *to see the pet*, with reminders seen incidentally.

Tiimo (visual daily planner; Apple "app of the year" laurels) shows a severe divergence between its 4.6★ cumulative rating and its recent reviews, 28.6% of which are one-star; its highest-concentration complaint cluster is notifications, and it barely exists on Android. **Habitica** (RPG-style gamification, 5M+ Play installs) demonstrates the punishment problem this project's red lines answer — its angriest reviews concern losing levels to a boss fight — and a years-long, still-open wound around Android notifications failing under battery management or arriving in overwhelming batches. **Numo** (positions itself as the non-infantile ADHD app) holds the positioning this project also targets but is burning it: 29% of its reviews concern billing disputes, and its Android rating has collapsed to 3.29★. A newer cluster of quest-style apps (Hyper, TaskHero, LifeUp) competes for the same users with "levelling up, not being told what to do".

Three design consequences: the pet-bond mechanism is validated and its ceiling untouched; data loss is existential for an emotional product and *worse* for PetTodo, whose pet is irreplaceable and whose storage is local-only; and subscription dark patterns are a category-wide trust failure that a competitor can differentiate against simply by not committing them.

1.3.3 The evidence on the design premises

Before this report I treated the anchor mechanism as established and the safety rules as obviously sufficient. I ran a structured literature search across sixteen questions to check both (method in §2.2); it changed my position on each, and the report states the revised positions rather than defending the original ones.

The anchor mechanism is not an established finding. The only controlled test of "body doubling" — working alongside another person — was null on 26 participants (Schuenke et al. 2025), and the broader social-facilitation meta-analysis puts mere presence at 0.3–3% of variance while finding it *impairs* complex-task performance (Bond & Titus 1983). Animal care has never been tested as a task-initiation aid in ADHD adults. Nothing I retrieved supports the chain *human presence helps* → *animal responsibility helps* → *a photo-derived virtual animal helps*. PetTodo therefore **tests** a mechanism rather than implementing a demonstrated one, and this report claims accordingly.

The rejected comparator works. If-then implementation planning improves inhibition, shifting and distraction resistance in children with diagnosed ADHD (Gawrilow & Gollwitzer 2008), on top of a large general-population meta-analysis (Gollwitzer & Sheeran 2006). "Reminders do not work, anchors do" inverts the evidential ordering.

Removing punishment does not remove guilt. Guilt is a documented post-abandonment state in self-trackers with no character, no failure state and no punitive feedback — 16.2% of activity-tracker users (Epstein et al. 2016) — because it is generated by the user's own appraisal. Anthropomorphic framing independently raises the felt cost of abandoning an entity (Chandler & Schwarz 2010), and an analysis of 582 r/Replika posts found harms mediated by role-taking: users felt the character had needs they were obliged to attend to (Laestadius et al. 2024). The zero-punishment red line is necessary but insufficient; the *shame reminder* finding says the same from the user's side.

The one experiment that manipulated pet feedback valence found warmth-only inert. Adolescents whose virtual pet gave both positive and negative feedback were roughly twice as likely to eat breakfast; the positive-only pet did not beat control (Byrne et al. 2012). The sample is adolescent, the outcome is breakfast, and guilt was not measured — but it is the closest existing test of this project's central safety rule, and it is not reassuring. I retain the rule as an ethical commitment; the evaluation must be able to see whether warmth-only is *sufficient*.

Three claims I had been repeating are withdrawn. The "prefrontal-to-amygdala switch" account of urgency-driven initiation does not survive contact with its sources — the finding concerns acute uncontrollable stress producing *impairment*, largely in animals (Arnsten 2015), while ADHD prefrontal regions are *under*-active. The "5–10% transfer rate" for single-mechanism ADHD interventions does not exist in the literature and is demoted to attributed practitioner opinion. And the 1–7 task cap cannot cite Miller: "seven plus or minus two" concerns immediate memory span, and Cowan describes the seven as a rhetorical device (Cowan 2001). The cap is instead defended by the effective-range observation in §1.3.2: small daily maintenance is where the pet mechanism works at all.

What the evidence does support. Warmth-only *reinforcement* has a direct ADHD result: on an incentive go/no-go task, participants with ADHD gained more from social reward — positive facial expressions — than controls (Kohls et al. 2009). And the event-based cue is the best-grounded interaction choice available, per the prospective-memory dissociation in §1.1. Notably, this supports the cue type independently proposed by the early-years special-needs specialist whose feedback my supervisor forwarded — a concrete relative anchor ("can we finish this before they finish making your lunch") — rather than the clock-triggered daily prompt I currently implement.

2. Methodology

2.1 Engineering method

Platform. I chose Flutter over two native codebases. The justification is scope: evaluation needs real users on whatever handset they own, and a solo project cannot maintain two native implementations to that standard. The cost is accepted deliberately — surfaces Flutter does not reach (iOS widgets, Android overlays) must be written natively, which is why the ambient layer is scheduled as separate work.

Architecture. I fixed layer boundaries before features. `lib/domain/` is pure Dart with no Flutter imports and owns the invariants — the one-to-seven rule, local-day rollover, positive-history aggregation, unlock thresholds; `lib/data/` owns storage, notification scheduling and export; `lib/sprite/` owns atlas parsing and rendering; `lib/application/` is a presentation-independent coordinator deliberately outside `lib/ui/`. A visual reskin therefore cannot move persistence or behaviour — demonstrated when the approved design was applied without touching domain logic — and the invariants are unit-testable without a widget harness.

Rendering. A small `CustomPainter` driven by a `Ticker`, rather than the Flame game engine: a game-engine dependency for one animated actor is disproportionate and would relocate animation state out of the tested layer.

Data. Everything is local. State is a versioned JSON document written by flush-then-rename; events are append-and-flush newline-delimited JSON. There is no account, no server and no analytics service. The justification is partly ethical — the community research records explicit refusal to hand mental-health-adjacent data to small developers' cloud services — and partly methodological, since it makes evaluation possible without data infrastructure. §5 records the cost: retention cannot be measured the way any cited study measures it.

Process. I develop against written task specifications, each stating goal, file scope, constraints and acceptance criteria before implementation begins, and I accept a change only after reading the full diff and running the build and test suite locally; user-visible changes are additionally walked through on a device or simulator before they land. Every feature must state which ADHD problem it solves; if that mapping cannot be stated, the feature is not built. This constitution is written down and version-controlled alongside the code, and several of its rules are enforced by tests rather than by discipline (§5).

2.2 Research method

Four bodies of evidence were gathered, deliberately of different kinds so each could check the others.

Community accounts. I read high-engagement threads and their top comments across three ADHD subreddits over two sweeps — one on tool abandonment, one on the condition's daily texture — recording per-thread links and vote counts so every claim remains traceable.

Store reviews. I collected 4,754 reviews via the public Apple review feed (1,154: Finch 400, Tiimo 350, Numo 254, Habitica 150) and a Play scraper (3,600: Finch 1,612, Habitica 1,537, Numo 339, Tiimo 112), deliberately oversampling one- to three-star reviews on the Play side. That choice is recorded next to every figure it affects: Play proportions indicate the *concentration* of a complaint, not its population prevalence.

Literature. I ran a structured search across sixteen questions covering the mechanism, the safety rules, and claims I had been repeating, treating contradicting evidence as the most valuable output and recording "no credible evidence found" where that was the truth rather than substituting adjacent work. The result is a 230-row citation table recording, per row, study design, sample, population mismatch, whether the claim is supported, and retraction flags checked against both OpenAlex and Crossref (two flagged, both disclosed). I additionally verified eight design-critical DOIs directly against the Crossref API — all eight resolved with matching titles — and checked the Altgassen finding line by line against the published abstract.

Limitations, stated up front. All community strands sample self-selecting, self-identifying, predominantly English-speaking online populations; store reviews are written by people still using a product, Reddit largely by people who left. Self-identification is itself a methodological problem: executive function rating scales and performance tests dissociate (median $r = .19$, [Toplak et al. 2013](#)), so case-control effect sizes

cannot be transferred to a self-identifying sample — only the weaker claim that self-rated difficulty predicts real functional impairment ([Barkley & Murphy 2010](#)).

3. Implementation

Core loop. Tasks, sprite engine, unlocks, local notifications and the JSONL event log, on both platforms.

Design and accessibility. The approved visual design plus a semantics rework — custom tap targets own container semantics above their gesture handlers, decorative art is excluded from the accessibility tree, onboarding scrolls on short viewports.

Raising system. State schema v2 with in-place migration, growth stages derived rather than persisted, a treat economy, a full-day pet schedule, long-press interaction, and a collection gallery with no progress bars.

To-do baseline. Schema v3. Tasks become ID-addressed objects of kind `daily` or `oneOff`, capped at one to seven; quick capture requires only a title; one-off tasks never carry age or overdue data. Positive history derives only from completion events and renders only weeks containing completions — no streak, gap, zero or missed-day state exists anywhere in the product.

Stability. A quick-capture crash on dialog dismissal, and an Android release build that black-screened because release resource shrinking stripped the notification icon — the fix keeps notification failures from ever blocking startup.

Own-pet hatch loop. The differentiating feature. The user photographs their animal; the app assembles a request package; a sprite pack (`.pettodopet`) produced by the image-generation pipeline is imported and installed atomically into application documents, and a runtime registry merges bundled pets with installed packs. A later change made onboarding list the live registry instead of a hardcoded single pet, and cleaned up exported request archives that previously accumulated indefinitely.

Animation quality. The idle animation read as choppy, something I noticed in my own daily use of the shipped build — I am inside the target population, which the evaluation treats as a stated limitation rather than a secret. Measurement located the cause: the six idle frames are generated independently, so 37–46% of pixels change between neighbouring frames and the silhouette drifts about 4 px — six similar dogs rather than one dog breathing. The fix holds a still frame while the pet rests, on the reasoning that the sprite contract itself describes idle as a "low-distraction" loop, and that a resting companion should be restful.

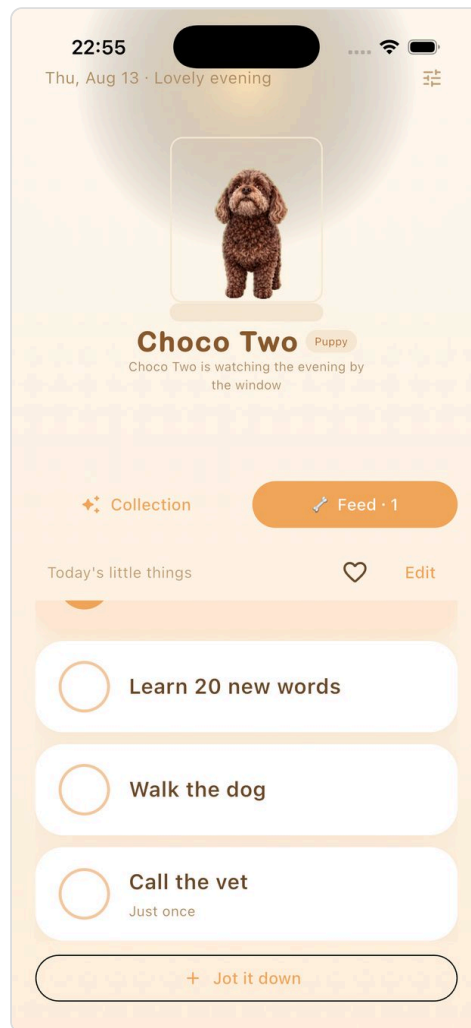
In progress, not yet part of the application. A reworking of the notification layer — new copy in the pet's voice reporting its own day, randomised selection, timing jitter, and back-off after unopened days. I am holding it unmerged because the literature contradicts two of its three mechanisms (§6).

4. Results

4.1 The working application

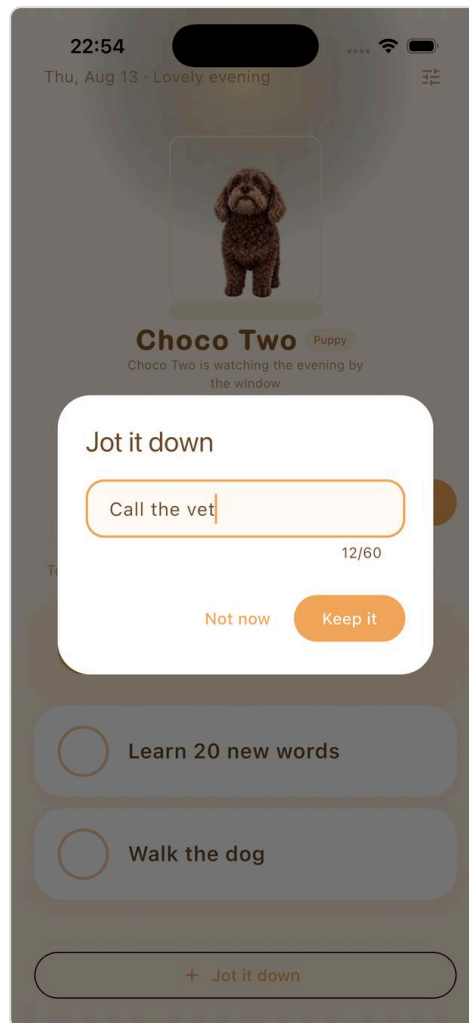
All screenshots below are from one continuous session on an iPhone 17 Pro running the current build, using a pet the user hatched from their own photographs. They are ordered as a user would meet them.

Figure 1 — Home with a mixed task list.



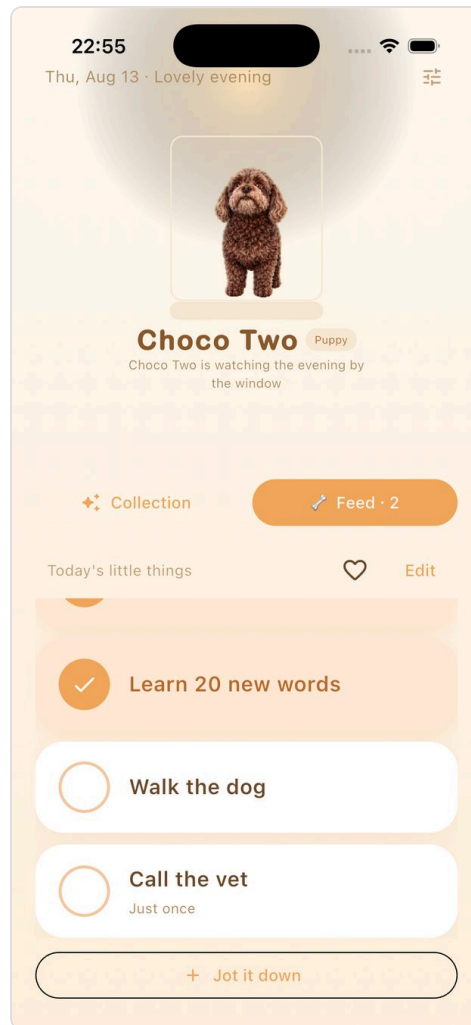
The pet occupies the top half of the screen and the task list the bottom: the proportion is the argument. The pet is not an ornament attached to a to-do list, it is what the screen is about, and the tasks are what a user encounters on the way to it. The list shows both task kinds — three recurring items and *Call the vet*, marked "Just once" — and the distinction is carried by that quiet subtitle alone. There is deliberately no due date, no age, no overdue marker and no count of what remains; a one-off task captured weeks ago looks exactly like one captured this morning. The task area is a bounded scrolling region rather than an expanding list, so the maximum of seven items can never grow into a full-screen wall. The header reads "Today's little things", and the completed item stays visible in warm tint rather than disappearing or being struck through.

Figure 2 — Quick capture.



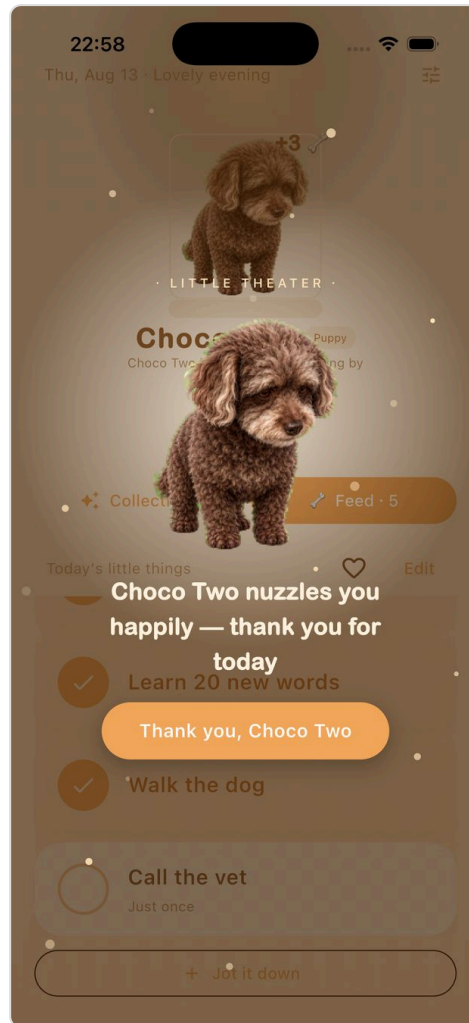
Capture is two steps: tap *Jot it down*, type, confirm. The field is autofocused, only a title is required, and the placeholder — "A thought before it slips away" — names the problem it solves. This is the ADHD symptom with the shortest fuse: a thought that leaves working memory in seconds cannot survive a form. Anything captured this way defaults to a one-off, because forcing a recurrence decision at capture time is exactly the friction that loses the thought. The dismissal option is worded "Not now" rather than "Cancel" or "Discard".

Figure 3 — Completing a task.



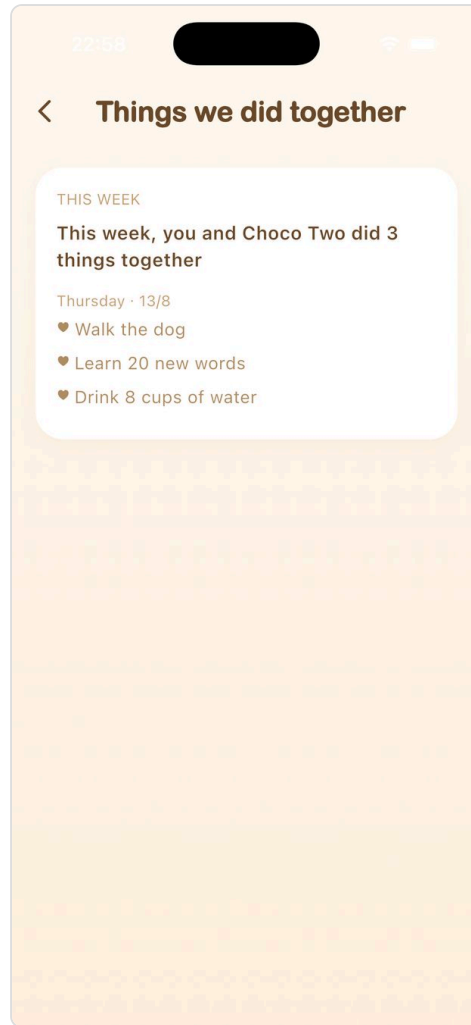
Completion produces warmth and nothing else: the card takes a warm tint, the circle fills, the pet plays a brief happy animation, and a treat drops — the counter has gone from one to two. There is no score, no streak, no progress bar and no "3 of 4 done" summary anywhere on the screen, because a progress indicator is also a deficit indicator. The treat is the only currency, it is spent on feeding the pet, and spending it is optional.

Figure 4 — The Little Theater, shown when the day's list is finished.



Finishing everything on the list triggers the one moment the application makes a fuss about: the pet fills the screen, particles rise, three bonus treats are awarded, and the message reads "Choco Two nuzzles you happily — thank you for today". The dismiss button says "Thank you, Choco Two", which keeps the exchange between the user and the animal rather than between the user and a scoreboard. This is the emotional peak of the design and the moment the whole reward loop exists to reach. It is also, by construction, the *only* moment with this weight — there is no equivalent screen for failure, because no failure state exists.

Figure 5 — Positive history.



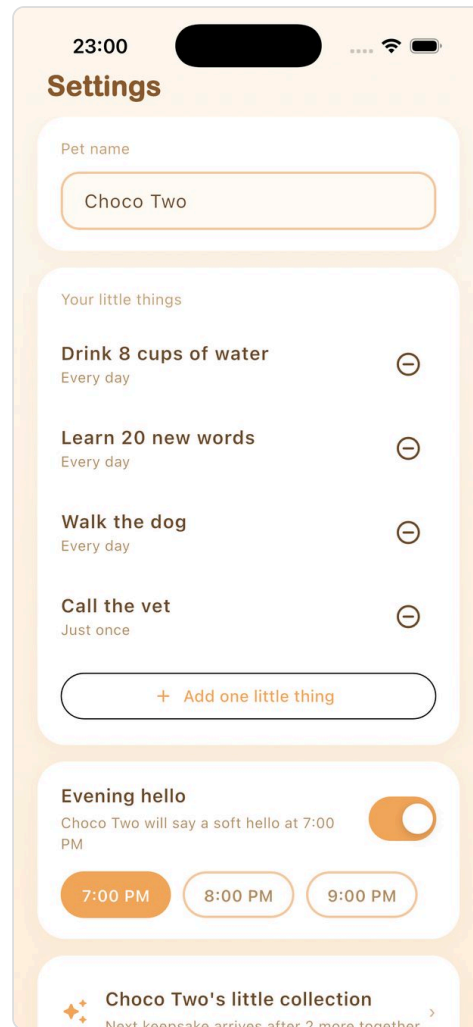
The history screen is the clearest single expression of the safety architecture. It reads "This week, you and Choco Two did 3 things together", and lists only Thursday, because Thursday is the only day with completions. Days without completions are not rendered as empty, greyed, or zero — **they do not exist in the data model at all**, so no view can accidentally surface them. There is no streak counter, no calendar grid with gaps to read as failure, and no comparison against a previous week. The framing is "things we did together", not "your completion rate".

Figure 6 — The collection.



Two things share this screen. The pet shelf shows both the bundled pet and the user's own hatched pet, with the active one outlined — the runtime registry described in §3 made visible, and the point at which the differentiating feature becomes tangible to the user. Below it, unlocked keepsakes would appear as illustrated items; locked ones show a paw silhouette and the words "A little mystery". They carry no progress bar, no unlock threshold and no "2 more to go", so the gallery cannot be read as a list of things not yet earned.

Figure 7 — Settings.



Settings is deliberately short. Tasks are listed with their kind stated in words ("Every day" against "Just once") and removed with a single control; there is no archive, no completed-items list and nowhere for finished work to accumulate. The notification section is one toggle and three fixed times, and its subtitle is phrased as something the pet does rather than something the user must configure. Notifications are off until explicitly enabled, and a denied permission is never requested again.

4.2 The hatch loop, end to end

The acceptance walkthrough for the differentiating feature was performed on the same device from a clean install: onboarding → hatchery → photograph → export of a 139 KB request archive → import through the real iOS document picker → both pets listed with the new one selected → switching between them → completing onboarding → Home showing the imported pet with its hatch ceremony — ending with zero leftover archives in documents. Two further screenshots taken during that run record the hatch ceremony itself and the resulting Home screen.

4.3 Implementation findings

Three findings from building it are results in their own right. A custom file extension is unpickable in the iOS document picker unless the app declares it: the picker produces a dynamic UTI and Files greys the file out, fixed by declaring `UTExportedTypeDeclarations`. Re-importing a pack for the currently selected pet paints a disposed image during the asynchronous gap unless the replacement atlas is loaded before the old one is disposed. And exported request archives accumulated indefinitely because cancellation deleted the request folder but not the archive beside it — found by inspecting the simulator's container, where the previous day's 139 KB archive was still sitting.

4.4 A negative result: the pixel-art experiment

The fourth result is negative and reported as such. To test whether a pixel-art style would improve inter-frame consistency, I first generated a pixel version through the sprite pipeline's own pixel preset: the output was not pixel art — 13,713 unique colours against fewer than 32 for the genuine article. Mechanical post-processing (downsampling plus palette quantisation) did produce true 23-colour pixel art but destroyed recognisability, because the eyes and muzzle are defined by exactly the detail downsampling removes first. What worked was constraining generation directly — a 64×64 logical grid, blocks aligned to it, a 24-colour palette, no anti-aliasing — which produced genuine pixel art that kept the identity anchors (the silver-brown muzzle, the brow spots, the floppy ear silhouette). A six-frame idle strip built from that reference measured 14% inter-frame drift after quantisation, against 42% for the current style, with frame area stable to within 0.2 percentage points: the silhouette stops breathing in and out between frames. The improvement is real but partial — 14% is still far above what a hand-animated loop would show — and the aesthetic choice it enables also answers the "infantile" objection documented against the category leader in §1.3.2.

5. Testing and evaluation

5.1 Software testing

`flutter test` passes 47 tests across 16 files — a figure I verified by running the suite, not by quoting it; the notification rework adds five more, including a lint that fails the build if invitation copy ever regains a forbidden phrasing. Coverage concentrates on the invariants the product promises its users: schema `v1→v2→v3` migration; day rollover, including that rolling over multiple missed days leaves no historical markers; one-off isolation from daily rollover; treat and unlock edges; the sorted, capped notification window with one fire per task per day and no re-asking after permission denial; local-week positive-history aggregation; JSONL round-trips; sprite atlas frame mathematics; pack install and validation; and accessibility semantics boundaries. The intent is that the red lines are enforced by tests rather than by discipline.

Gaps before submission: no end-to-end automated UI test; the ambient layer will need platform-specific test provision; and Android notification timing needs device-level verification, since the review corpus shows scheduled notifications failing or arriving in batches under OEM battery policies to be a category-wide defect (111 of 1,537 Habitica Play reviews, 63 of them one or two stars).

5.2 User evaluation

The full plan is written up separately and summarised here: five to ten seed participants who are ADHD-leaning pet owners, recruited through my personal network only; a fourteen-day run; a short questionnaire at days 7 and 14; user-initiated local data export as the only data channel; and qualitative red-light monitoring for guilt or uncanny reactions to the user's own animal, treated as a finding that outweighs any retention number. Covert recruitment in ADHD communities is excluded both ethically and pragmatically — the community research documents moderators publicly naming products for stealth marketing. Ethics approval is the longest-lead-time item and the first question for the school.

Two revisions to the evaluation follow from the literature and are recorded here because they change what the evaluation can claim. First, the community benchmark of ">50% retention at two weeks" turns out to conflate two incompatible frames: real-world median 15-day retention for mental-health apps is 3.9% (Baumel et al. 2019), while recruited trials run near 75% and report usage a median 4.06× higher than real-world use of the same programs (Baumel et al. 2019). A recruited seed group is a trial population, so >50% is a weak bar in that frame and an extraordinary one in the other; the report will state which frame it is in. Second, two weeks is the wrong window anyway: the review corpus locates two failure points, days 2–4 when setup enthusiasm fades and roughly the second month when novelty does. A day-60 check is added. Unblinded self-report is additionally the most inflation-prone outcome in this field — behavioural-intervention effects in children collapsed to non-significance under probably-blinded assessment (Sonuga-Barke et al. 2013) — which the analysis acknowledges by privileging the logged event data over questionnaire answers wherever the two disagree.

6. Conclusion and future work

At mid-term the application does what it was specified to do, and the research conducted for this report says the specification was partly wrong. That is the honest summary, and the remaining work follows from it rather than from the original plan.

Resolve the notification design against the evidence. The unmerged branch implements randomised copy, timing jitter, and exponential back-off after unopened days. The literature supports none of the three as mechanisms: the closest test found a purpose-built 30-message bank no better than a single fixed string (Bell et al. 2023); jitter is opposed by cue-consistency accounts of habit formation (Lally et al. 2010); and one trial found *less* frequent notification produced less viewing and actioning (Morrison et al. 2017). Worse, withdrawing contact is ambiguous by construction, and rejection sensitivity is defined as readily perceiving intentional rejection in ambiguous behaviour (Downey & Feldman 1996) — so back-off may be

a guilt pathway rather than a protection. The copy rewrite is retained on separate grounds; the two mechanisms are under review.

Move from time-based to event-based cues. The clearest evidence-led change available and not yet built: anchoring a task to "after dinner" rather than to 19:00 uses the prospective-memory channel that is spared in ADHD adults rather than the one that is impaired, and matches the cue type the forwarded expert feedback independently proposed.

Design the return screen and an explicit pause. Zero punishment does not prevent an accumulated backlog from speaking — the *shame reminder*. Two mechanisms answer it: a first screen after absence that shows the pet's own accumulated news rather than the user's arrears, and a deliberate pause action that reframes absence as chosen rather than failed.

Promote export to a user-facing backup. The application is local-only, the pet is generated from the user's own animal and is therefore irreplaceable, and data loss is the top-voted complaint against the category leader — which at least has accounts to restore from. The export mechanism exists; making it a visible backup with an honest explanation is a small change against a disproportionate risk.

Finish the style decision, then build the ambient layer. Pixel art is technically viable (§4), and the ambient layer has the strongest single piece of user evidence in the review corpus — a five-star review describing the widget's pet speaking on a day the user had already given up. It is sequenced after the style decision because the asset form determines what the widget draws, and it will not be built before the Android battery-policy and notification-batching risks are addressed.

Evaluation. Ethics enquiry, recruitment, the fourteen-day run with a day-60 follow-up, and analysis. The outstanding intellectual work is deciding what the project claims when its central mechanism is, on present evidence, unproven — and running an evaluation honest enough to find out.